The effect of retirement on firms and co-workers

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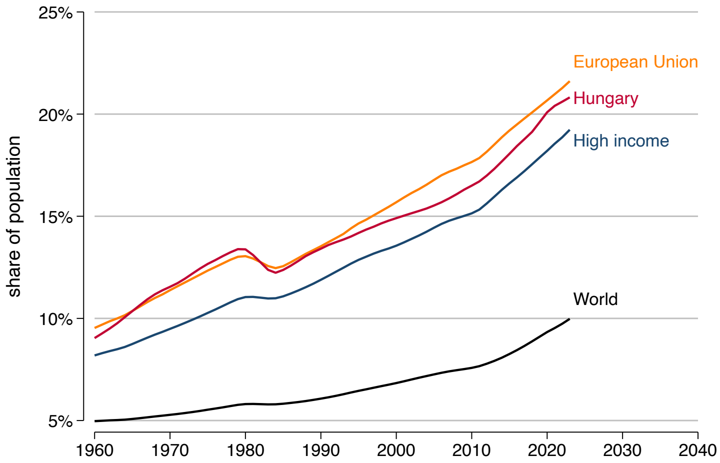
Society is aging

"By 2030, 1 in 6 people in the world will be aged 60 years or over." (WHO, 2025)

Share of population ages 65 and above (World Bank)



This means...

- Firms struggle with older workforce
- Each year a larger share of workers exit the labour market because of retirement
  - workers with high-level human capital
  - decreasing productivity because of decreasing health and human capital
- **Research question:** What is the impact of retirement on firms and co-workers?

This paper

- Estimate the effect of worker retirement on **firm-level** and worker-level **outcomes**
- Using Hungarian administrative data
- Identification: DiD and IV based on a unique early retirement policy change
- **I find that...**
  - retirement increases labour productivity
  - firms cannot/do not want to replace retired workers

Related literature

- **Worker exit**
  - High hiring and training costs (firm specific human capital) (Abowd and Kramarz, 2003; Muehlemann and Pfeifer, 2016; Becker, 1962; Bartel et al., 2014)
  - External hires are not perfect substitutes (Jäger et al., 2024; Jaravel et al., 2018; Badalyan, 2025)
  - Spillover effects on incumbent workers (Azoulay et al., 2010; Jäger et al., 2024; Jaravel et al., 2018; Waldinger, 2012)
  - **Contribution:** Estimate the effect of the exit of a specific group, older workforce
- **Pension related policy changes**
  - General or early retirement age changes (Simonovits and Tir, 2017; Bíró and Elek, 2018; Bianchi et al., 2023; Carta et al., 2021; Boeri et al., 2022).
  - **Contribution:** Use unique a policy change, that decreases early retirement age
- **Productivity of older workers**
  - Diminishing marginal productivity, less investment into human capital (Cardoso et al., 2011; Skirbekk, 2004)
  - High wages relative to productivity (Lazear, 1979)
  - **Contribution:** Estimate the contribution of older workers to overall firm productivity

Institutional settings

- Mandatory pay-as-you-go system: benefits are based on earnings, with a minimum pension
- Different pension calculation system after 2007 (8% smaller starting pension Simonovits and Tir (2017))
- Eligibility for pension: min. 20 years of service
- Worker protection in last 5 years before retirement age
- Until 2012, two statutory retirement age levels
    - General retirement age: 62
    - Early retirement age:
        - Men: 60
        - Women: 57 $\rightarrow$ 59 (from 1952 cohort)
- 2012: abolition of early retirement option, except Nők40 (Women40)

Women40

2011: introduction of Women40

- Announcement in November 2010, introduction in January 2011
- Women with 40 years of contribution can retire earlier
- 32 years of working relationship
- High incentive to retire earlier: no reduction in pension benefits with relatively high aggregate replacement ratio (Eurostat (2025): HU: 60%, EU: 56%)
- People retire as soon as they become eligible for early retirement (even with punishment in benefit)
- Reduced median women retirement age by 2.2 years (Simonovits and Tir, 2017)

Data

ADMIN3: linked employer-employee administrative data

- 2003-2017
- 50% sample of the Hungarian population in 2003
- Covers approximately 5 million individuals
- Monthly level employment and pension transfer, and pension determination time

Retirement definition:

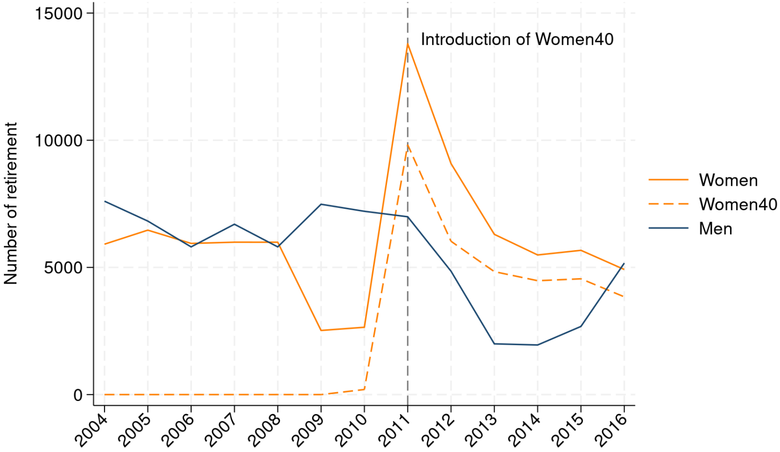
- Main workplace with normal work contract
- Last workplace before pension determination
- Continuous workplace for 6 months before exit
- The individual do not work for this firm anymore in the data

Data

Merge detailed firm characteristics using data from the Hungarian Tax Authority

- Administrative tax declaration forms
- Employ at least one worker anytime in the ADMIN3 dataset in 2003-2017
- More than 300K firms yearly
- Balance sheet, income statements, industry, employment

Yearly number of retirement observed in ADMIN3



Notes: Agriculture and non-market services are excluded. Retired workers excluded after 2007.

Retirement by type

Main sample

- Exclude agriculture and non-market service industries
- Keep only 2008-2013 period:
  - before 2008: cannot see if someone is working while receiving pension
  - after 2013: increased general retirement age Cohorts
- observe at least 3 employees in ADMIN3 in 2010
- 5-50 employees in 2010 from balance sheet data (Badalyan, 2025; Jäger et al., 2024; Jaravel et al., 2018):
  - effect of retirement decreases with size: hard to detect it in larger firms
- Control group: no retirement, but have at least 1 old worker (50+) in 2008-2013
- Treatment group: retirement only in 2011 and only 1 worker retired and have at least 1 old worker (50+) in 2008-2010

Yearly frequency of firms

Year	Raw	Main sample	Control	Treatment
2008	307,206	9,164	8,107	1,057
2009	319,055	9,164	8,107	1,057
2010	330,915	9,164	8,107	1,057
2011	352,328	9,164	8,107	1,057
2012	350,955	9,082	8,107	975
2013	350,942	9,045	8,107	938
Obs.	**2,011,401**	**54,783**	**48,642**	**6,141**

Comparison of control and treated group in 2010

Variable	**Control**	**Treated**	**t-stat.**
Number of workers	16.98	18.93	-4.95***
	(12.00)	(12.23)	
ln(Value added)	10.74	10.73	0.39
	(1.08)	(1.09)	
ln(Value added/worker)	8.11	7.97	4.79***
	(0.86)	(0.87)	
Avg. worker age	43.99	44.99	-5.62***
	(5.46)	(5.28)	

Notes: Standard deviations in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Empirical strategy

$$y_{it} = \sum_{t \neq 2010} \beta_t^{HS} \left(\text{HS}_i \times \mathbf{1}\{\text{Year} = t\}\right) + \sum_{t \neq 2010} \beta_t^{LS} \left(\text{LS}_i \times \mathbf{1}\{\text{Year} = t\}\right) + \alpha_i + \mu_t + \varepsilon_{it}$$

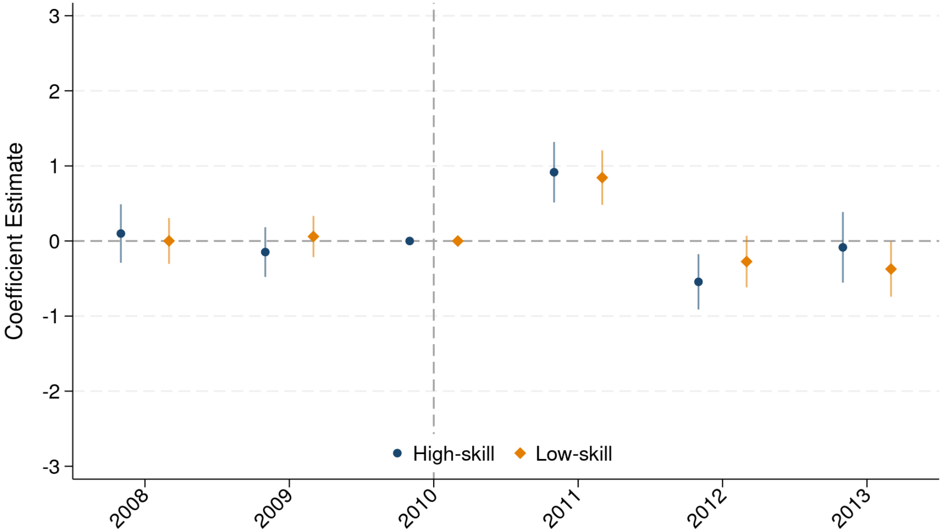
- $y_{i,t}$: number of worker exits and entries, ln(number of workers), ln(value added), and ln(value added/worker) in firm i at year t
- $\text{HS}_i = 1$ if a high-skilled worker retired from firm i in 2011
- $\text{LS}_i = 1$ if a low-skilled worker retired from firm i in 2011
- α_i, μ_t: Firm and time fixed effects

Worker Skill Classification

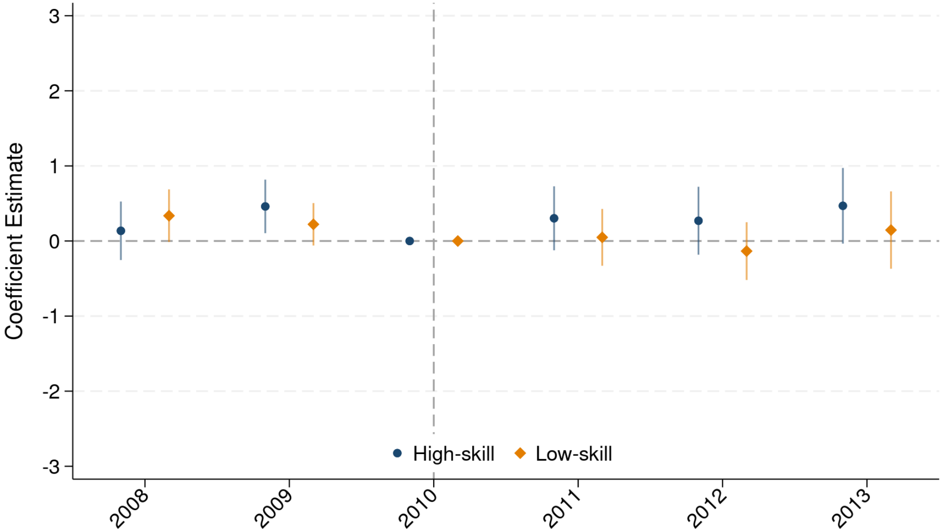
Based on 1-digit FEOR (ISCO) codes

- **High-skill workers**: Managers, Professionals, and Technicians (1-3 major groups)
- **Low-skill workers**: Clerical service and sales workers, Skilled agricultural and trades workers, Plant and machine operators, and assemblers, Elementary occupations (4-9 major groups)

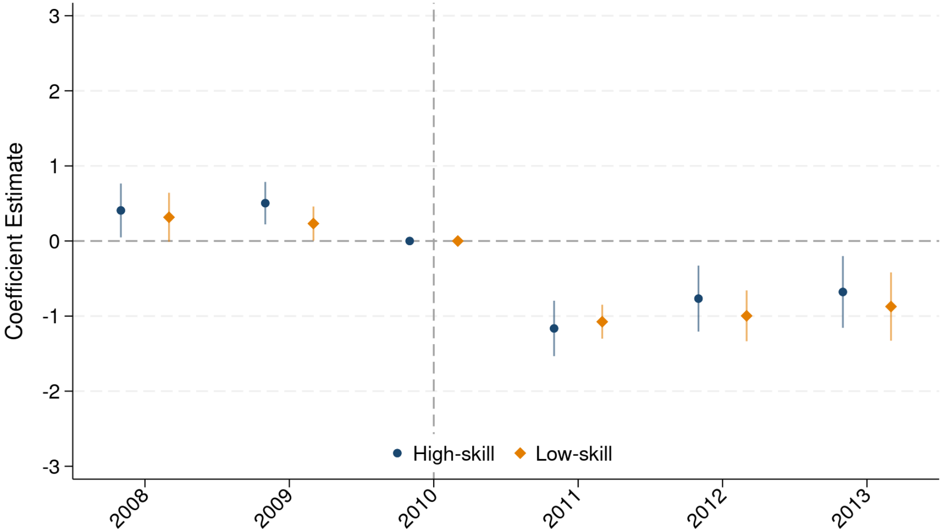
Event study - Exit



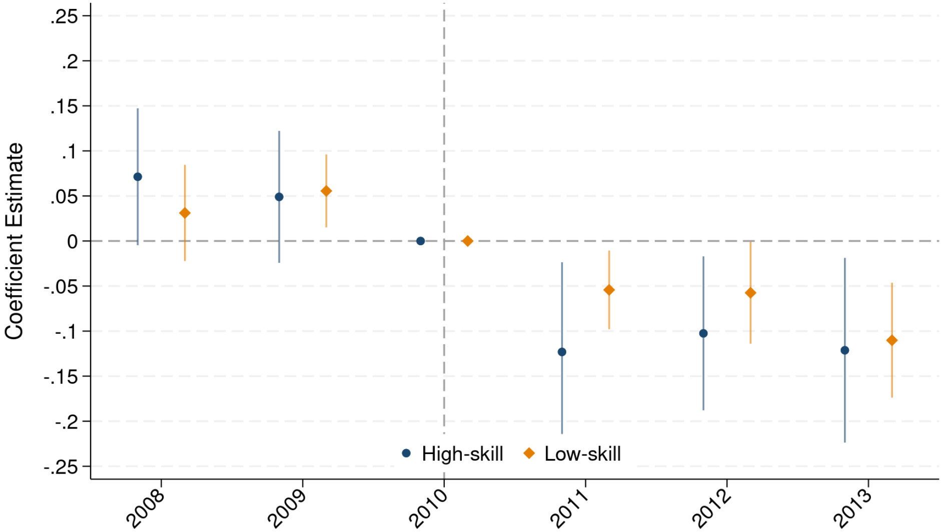
Event study - Entry



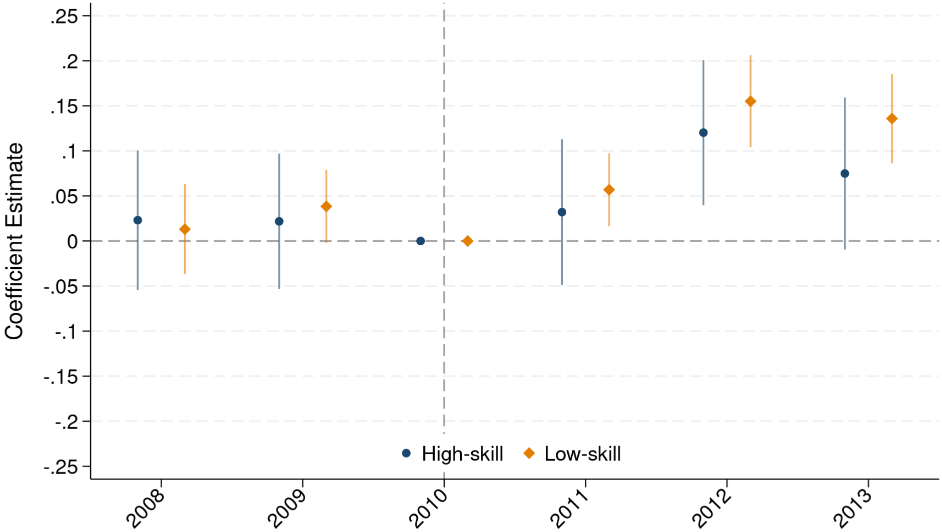
Event study - Workers



Event study - ln(Value added)



Event study - ln(Value added/worker)



Potential endogeneity

Retirement is not random

- selection bias: firms try retain better workers
- workers use retirement to "leave the sinking ship"
- firms layout older workers to cut costs

IV estimation

$$\Delta\left(y_{it}\right) = \alpha + \beta_{1IV}\Delta\hat{HS}_{it} + \beta_{2IV}\Delta\hat{LS}_{it} + \upsilon_{it}$$

First stage regressions

$$\Delta\hat{HS}_{it} = \pi_0 + \pi_1\Delta HSW40_{it} + \pi_2\Delta LSW40_{it} + \eta_{it}$$

$$\Delta\hat{LS}_{it} = \phi_0 + \phi_1\Delta HSW40_{it} + \phi_2\Delta LSW40_{it} + \zeta_{it}$$

- Δ changes from $t-1$ to t
- HS_{it}, LS_{it}: high-skilled and low-skilled worker retirement in 2011
- $HSW40_{it}, LSW40_{it}$: has high-skilled or low-skilled workers eligible for Women40 in 2011
- Women40 eligiblity: used Women40 to retire or 59-61 year old women in 2011

Identification assumptions

Relevance condition First Stage

- Women40 policy directly affects retirement behavior
- significant jump in retirement following the policy change
- first stage regression: instruments are strong and relevant
    - high-skill and low-skill instruments are statistically significant
    - F-statistics are above the common threshold of 10

Exclusion restriction

- Women40 policy impacts firm productivity only through worker exits
- policy change increases only women's retirement eligibility
- impact on firm productivity only through worker retirement

IV Results - Value added/worker

	FD			IV		
	Workers	ln(VA)	ln(VA/workers)	Workers	ln(VA)	ln(VA/workers)
ΔHS	-1.058***	-0.077	0.066*	-2.040*	0.064	0.147
	(0.185)	(0.050)	(0.040)	(1.095)	(0.195)	(0.203)
ΔLS	-0.982***	-0.021	0.083***	0.808	0.439***	0.270**
	(0.110)	(0.021)	(0.020)	(0.854)	(0.126)	(0.114)
N	45,470	43,840	43,753	45,470	43,840	43,753

Notes: Clustered standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Conclusion

- Retirement has a positive effect on firm productivity
  - Heterogeneity by skill of the retired worker
- There is still potential endogeneity
- IV results: Only LS retirement has a positive effect
  - but probably it is a weak IV

Retirement age by cohort

	General				Early			
	Women		Men		Women		Men	
Cohort	Age	Year	Age	Year	Age	Year	Age	Year
1947	62	2009	62	2009	57	2004	60	2007
1948	62	2010	62	2010	57	2005	60	2008
1949	62	2011	62	2011	57	2006	60	2009
1950	62	2012	62	2012	57	2007	60	2010
1951	62	2013	62	2013	57	2008	60	2011
1952/1	62.5	2014/2	62.5	2014/2	59	2011	60	2012
1952/2	62.5	2015/1	62.5	2015/1	59	2011	60	2012
1953	63	2016	63	2016	–	–	–	–

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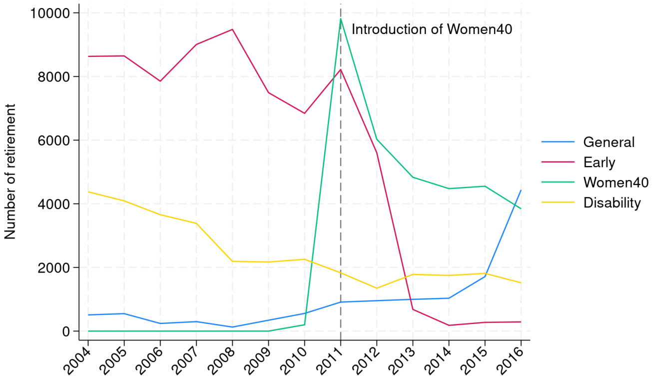
Yearly number of retirement by pension type



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First stage regressions

	ΔHS	ΔLS
ΔHS eligible	0.093***	0.029***
	(0.013)	(0.010)
ΔLS eligible	-0.001	0.121***
	(0.003)	(0.011)
F-statistic	26.37	64.68
Observations	43,753	43,753

Notes: Clustered standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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